

## **COURSE PROJECT REPORT**

ON

### **Evaluation Reliability of a Single Train–Test Split A Comparison with Repeated Random Splits and 5-Fold Cross-Validation**

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# Evaluation Reliability of a Single Train–Test Split

*A Comparison with Repeated Random Splits and 5-Fold Cross-Validation*

## 1. Research Question

**How reliable is a single train–test split as a performance estimate?**

The investigation examines whether the reported performance of a machine-learning model changes when the training and test data are partitioned differently, and compares a single 80/20 split with repeated random 80/20 splits and 5-fold stratified cross-validation.

## 2. Hypothesis

A single train–test split can provide a variable estimate of model performance because the measured test performance depends on the particular observations assigned to the training and test sets. Repeating the split should reveal this variability, while cross-validation should provide a more stable estimate by evaluating the model over multiple partitions.

## 3. Theoretical Background

A train–test split separates the available observations into a training set used to fit the model and a test set used to estimate performance on unseen observations. With only one split, the reported metric is conditional on that particular partition.

- **Repeated random splitting:** repeating the train–test procedure makes the distribution of performance estimates visible and allows variation to be quantified.
- **Cross-validation:** K-fold cross-validation partitions the data into K folds and uses a different fold for evaluation in each iteration, providing multiple performance estimates.
- **Evaluation reliability:** lower variation among repeated performance estimates indicates greater stability of the evaluation procedure.
- **Generalization:** test performance is used as an estimate of how the trained model behaves on unseen data.

## 4. Dataset / Data Source

The notebook uses the UCI Wine Quality *white wine* dataset (`winequality-white.csv`) [2]. It initially contains 4,898 rows and 12 columns; 937 duplicate rows and no rows with missing values are reported. After duplicate removal, 3,961 rows remain. Duplicates are removed before any splitting so that identical rows cannot appear in both the training and test sets.

Six features are selected for the experiment: alcohol, volatile acidity, density, chlorides, sulphates, and citric acid. The target variable is `quality`. The final feature matrix has 3,961 rows and 6 features, and the target vector has 3,961 observations.

| Dataset item                    | Value             |
|---------------------------------|-------------------|
| Original shape                  | $4,898 \times 12$ |
| Duplicate rows                  | 937               |
| Rows with missing values        | 0                 |
| Final shape after preprocessing | $3,961 \times 12$ |
| Selected features               | 6                 |
| Target                          | quality           |
| Final X shape                   | $3,961 \times 6$  |
| Final y shape                   | 3,961             |

**Suitability:** the dataset has enough observations for repeated stratified 80/20 splits and 5-fold cross-validation while keeping the model and features fixed, allowing the investigation to focus on evaluation variability.

## 5. Experimental Design

| Element             | Experimental choice                                        |
|---------------------|------------------------------------------------------------|
| Model               | RandomForestClassifier                                     |
| Number of trees     | 100                                                        |
| Model random state  | 42                                                         |
| Single split        | 80% training / 20% test, stratified (random_state = 42)    |
| Repeated splits     | 30 stratified random 80/20 splits (random_state = 1 to 30) |
| Cross-validation    | 5-fold StratifiedKfold, shuffled, random_state = 42        |
| Metrics             | Accuracy, weighted precision, weighted recall, weighted F1 |
| Additional measures | Generalization gap and computational time                  |

The investigation consists of three experimental conditions: (1) a single 80/20 train–test split; (2) thirty repeated stratified 80/20 splits; and (3) five-fold stratified cross-validation using the same model configuration. The controlled elements are the dataset, feature set, target, model family, number of trees, and model random state. The main factor changed is the way the data are partitioned and evaluated.

## 6. Results

The notebook provides numerical results for all three evaluation procedures. The single split produced one performance estimate, while the repeated-split and cross-validation experiments produced distributions of estimates from which the mean, standard deviation, minimum, maximum, and range were calculated.

## 6.1 Single 80/20 Train-Test Split

| Metric             | Result   |
|--------------------|----------|
| Accuracy           | 0.5233   |
| Weighted Precision | 0.4995   |
| Weighted Recall    | 0.5233   |
| Weighted F1 Score  | 0.4941   |
| Training Accuracy  | 1.0000   |
| Generalization Gap | 0.4767   |
| Computational Time | 0.8563 s |

## 6.2 Thirty Repeated Random 80/20 Splits

| Metric    | Mean     | Std. Dev. | Minimum  | Maximum  | Range    |
|-----------|----------|-----------|----------|----------|----------|
| Accuracy  | 0.529634 | 0.014620  | 0.503153 | 0.558638 | 0.055485 |
| Precision | 0.513535 | 0.021494  | 0.483894 | 0.571178 | 0.087284 |
| Recall    | 0.529634 | 0.014620  | 0.503153 | 0.558638 | 0.055485 |
| F1 Score  | 0.505411 | 0.013678  | 0.479507 | 0.533947 | 0.054440 |

Average generalization gap: 0.4704. Total computational time: 27.6240 s. Average time per split: 0.9208 s.

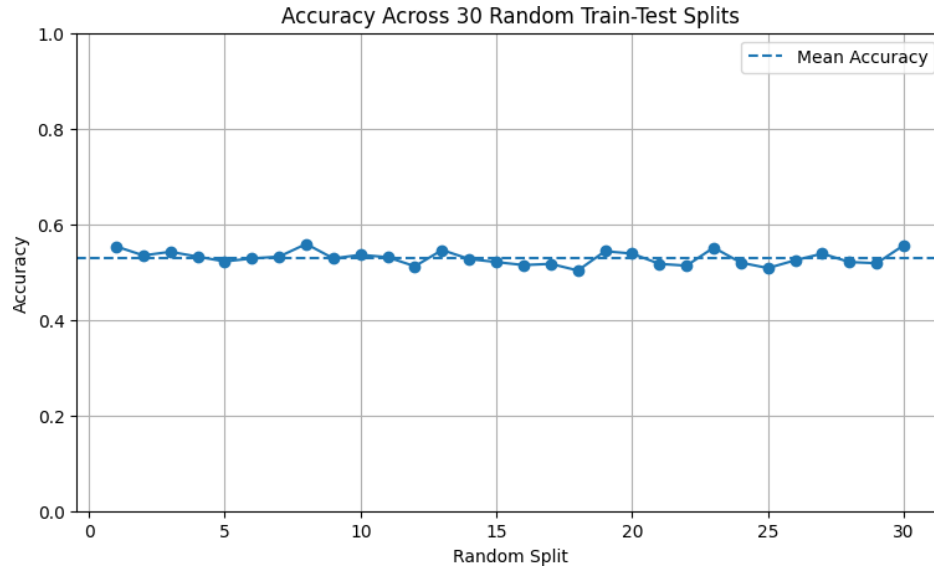


Figure 1. Accuracy across the 30 random train-test splits.

### 6.3 Five-Fold Cross-Validation

| Metric    | Mean     | Std. Dev. | Minimum  | Maximum  | Range    |
|-----------|----------|-----------|----------|----------|----------|
| Accuracy  | 0.530927 | 0.007032  | 0.525253 | 0.542929 | 0.017677 |
| Precision | 0.510632 | 0.012193  | 0.494973 | 0.524591 | 0.029618 |
| Recall    | 0.530927 | 0.007032  | 0.525253 | 0.542929 | 0.017677 |
| F1 Score  | 0.505085 | 0.007259  | 0.496665 | 0.516249 | 0.019585 |

Average generalization gap: 0.4691. Total computational time: 4.2118 s. Average time per fold: 0.8424 s.

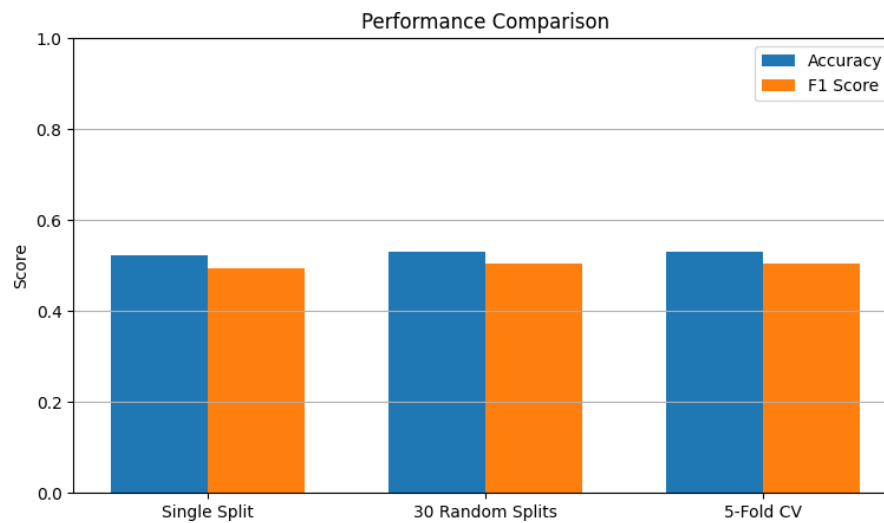


Figure 2. Mean performance comparison across the three evaluation methods.

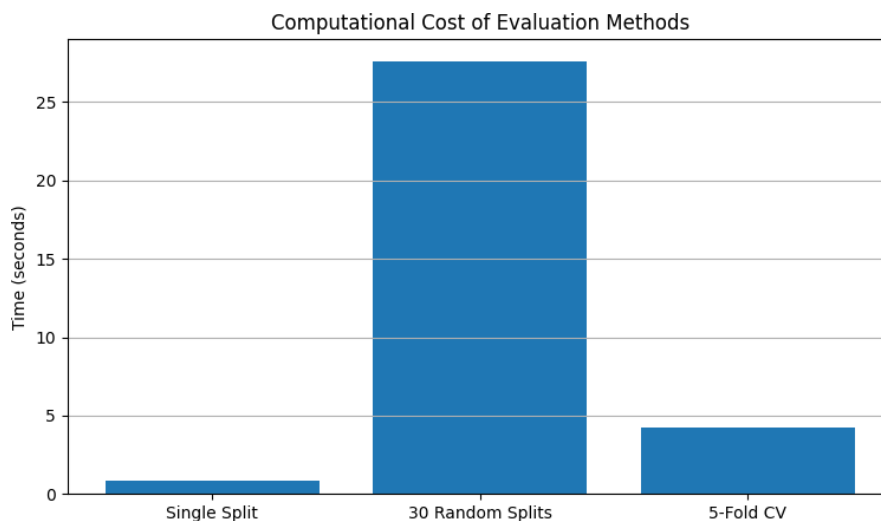


Figure 3. Computational cost of the three evaluation methods.

## 7. Analysis and Interpretation

The single split gave an accuracy of 0.5233. Across 30 stratified 80/20 partitions, mean accuracy was 0.5296 (SD 0.0146; range 0.0555), so the estimate varies depending on which split is selected. Precision varied the most across splits (range 0.0873), while accuracy, recall and F1 score each varied by roughly 0.055.

The single-split accuracy was approximately 0.0063 below the mean of the 30 repeated splits and approximately 0.0076 below the 5-fold cross-validation mean (0.5309). These differences are small in absolute terms, and the single-split value lies inside the range seen across the random splits (0.5032 to 0.5586). However, a single split represents only one point in a broader distribution, and a different random seed could have reported a noticeably different value.

Cross-validation showed lower variability in accuracy (SD 0.0070) than the repeated random splits (SD 0.0146), and its range was also smaller (0.0177 compared with 0.0555). In this experiment it therefore produced a tighter set of estimates. This comparison should be read with care: the 30 random splits overlap and the cross-validation SD is based on only five folds, so neither SD is a formal confidence interval.

The mean accuracies were close: 0.5233 (single split), 0.5296 (repeated splits) and 0.5309 (cross-validation). The main difference between the methods is therefore not the average level of performance but how much information they give about the variation of the estimate.

**Computational trade-off:** 0.8563 s for one split, 27.6240 s for 30 splits and 4.2118 s for 5-fold cross-validation. Repeating the evaluation 30 times cost roughly 32 times more than a single split, while cross-validation cost roughly 5 times more. Repeated evaluation costs more but shows how variable the estimate is.

Training accuracy was 1.0000 compared with a test accuracy of 0.5233 (a gap of 0.4767). The average gaps were 0.4704 (repeated splits) and 0.4691 (cross-validation). This large, stable gap reflects the Random Forest fitting the training data perfectly and is not a concern for the central reliability question.

### 7.1 Summary Comparison

| Evaluation method       | Mean Accuracy | Accuracy Std. Dev. | Accuracy Range | Time      |
|-------------------------|---------------|--------------------|----------------|-----------|
| Single 80/20 split      | 0.5233        | —                  | —              | 0.8563 s  |
| 30 random 80/20 splits  | 0.5296        | 0.0146             | 0.0555         | 27.6240 s |
| 5-fold cross-validation | 0.5309        | 0.0070             | 0.0177         | 4.2118 s  |

## 8. Critical Evaluation and Limitations

- The repeated-split experiment uses 30 random partitions. A different number of repetitions could give a somewhat different estimate of the variability.
- Only one single split (seed 42) is examined. A different seed would have produced a different single-split result, so the comparison concerns one particular draw.
- The experiment uses one dataset and one Random Forest configuration with no hyperparameter tuning. The observed behaviour should not automatically be generalized to other datasets or algorithms.
- The selected six features represent only one feature subset. A different feature selection could change model performance and the observed variability.
- Duplicate removal reduces the dataset from 4,898 to 3,961 rows. This preprocessing choice affects the absolute accuracy values and should be kept in mind when interpreting them.
- Precision, recall and F1 are weighted averages over several quality classes and can hide weak performance on rare classes.

- Computational times depend on the hardware and software environment in which the notebook was executed, so they are useful mainly for comparing the methods within the same run.

## 9. Conclusion

The investigation shows that a single train–test split provides only one performance estimate and does not reveal how much the estimate can vary across alternative partitions. In this experiment, the single split produced an accuracy of 0.5233, while 30 repeated random splits produced a mean of 0.5296 with a standard deviation of 0.0146 and a range of 0.0555.

Five-fold cross-validation produced a mean accuracy of 0.5309 with a smaller standard deviation of 0.0070 and a range of 0.0177. The repeated evaluation procedures therefore supplied additional evidence about variation that cannot be obtained from one split alone, at the price of additional computation.

For the dataset and model configuration used in this project, the evidence supports the conclusion that a single train–test split should be interpreted as one sample of possible performance estimates rather than as a complete description of evaluation variability. Repeated splitting and cross-validation give more information about the stability of the estimate, and the choice involves a trade-off between the information gained and the computational cost.

## 10. AI Usage Declaration

| Required item        | Declaration                                                                                                                                                                                                                 |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI tool(s) used      | ChatGPT; Claude (Anthropic)                                                                                                                                                                                                 |
| Purpose of using AI  | Assistance with report structure, code changes, debugging, restructuring the notebook documentation, and checking the notebook against the provided project requirements.                                                   |
| Parts assisted by AI | Report organization, interpretation drafts, notebook markdown sections, and code debugging. All results were checked against the executed notebook outputs, and the team reviewed the methodology, results and conclusions. |

## 11. References

- [1] St. Joseph's College of Engineering and Technology, "24SJPCST503 – Machine Learning: Course Project Guidelines," Academic Year 2026–2027.
- [2] P. Cortez, A. Cerdeira, F. Almeida, T. Matos and J. Reis, "Modeling wine preferences by data mining from physicochemical properties," *Decision Support Systems*, vol. 47, no. 4, pp. 547–553, 2009. Dataset: UCI Machine Learning Repository, Wine Quality (white wine), <https://archive.ics.uci.edu/ml/datasets/wine+quality>.